AWS Setup and Security Alerting Project

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Project Overview

This project involved setting up an AWS Free Tier environment to implement a real-time security alerting system focused on monitoring root user activity. The main goal was to detect and respond to potentially unauthorized use of the AWS root account by tracking the `GetCallerIdentity` API call—an action often used during reconnaissance or by attackers validating credentials. To achieve this, the project utilized key AWS services: **CloudTrail** for logging, **SNS (Simple Notification Service)** for notifications, and **EventBridge** for event-driven automation.

By integrating these tools, the system was able to generate immediate alerts via email when a sensitive action was detected, demonstrating an effective approach to enhancing cloud infrastructure security.

Objectives

Create an AWS Free Tier account to conduct experiments in the cloud.

Make sure CloudTrail is set up to record AWS API activities.

SNS (Simple Notification Service) from Amazon can be used to deliver email notifications.

Configure Amazon EventBridge so that root user actions will cause alerts to be sent.

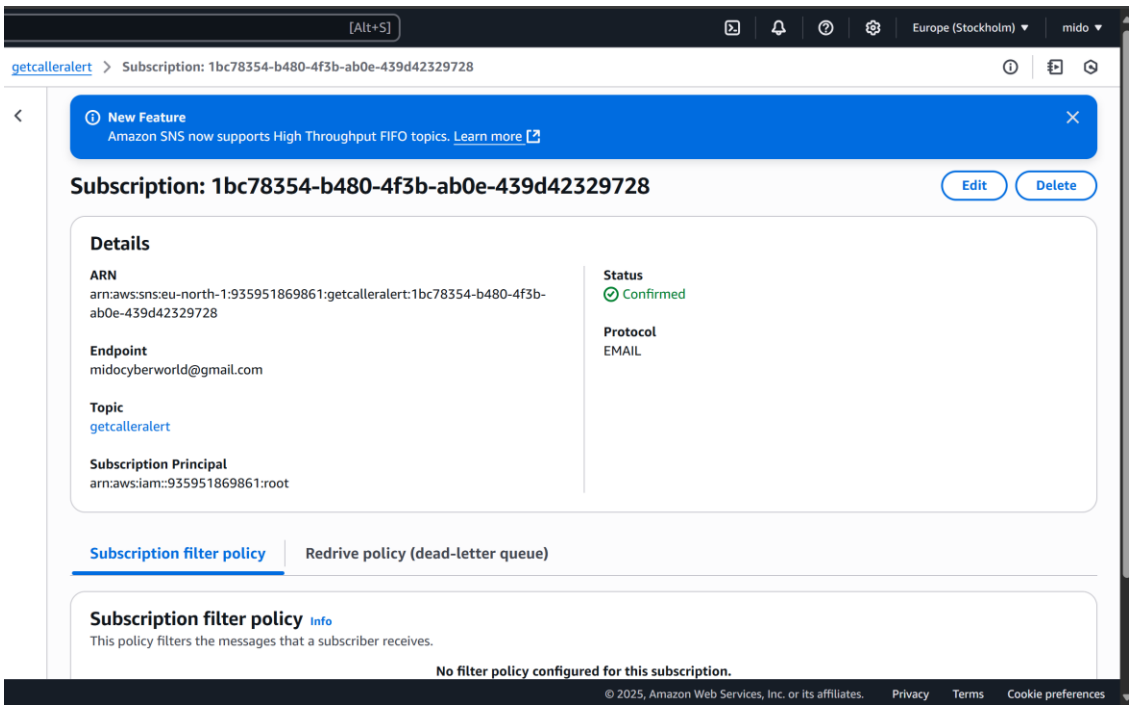
To verify functionality, use the AWS CLI to test the configuration.

Technologies Used

- **AWS CloudTrail**: Tracks and logs all AWS API requests made within the account.
- **Amazon SNS**: Publishes notifications to email subscribers based on triggered events.
- **Amazon EventBridge**: Routes specific API events to SNS for alerting.
- **Kali Linux CLI**: Used to simulate API calls and test alert functionality.
- **AWS CLI**: Command Line Interface tool for interacting with AWS services programmatically

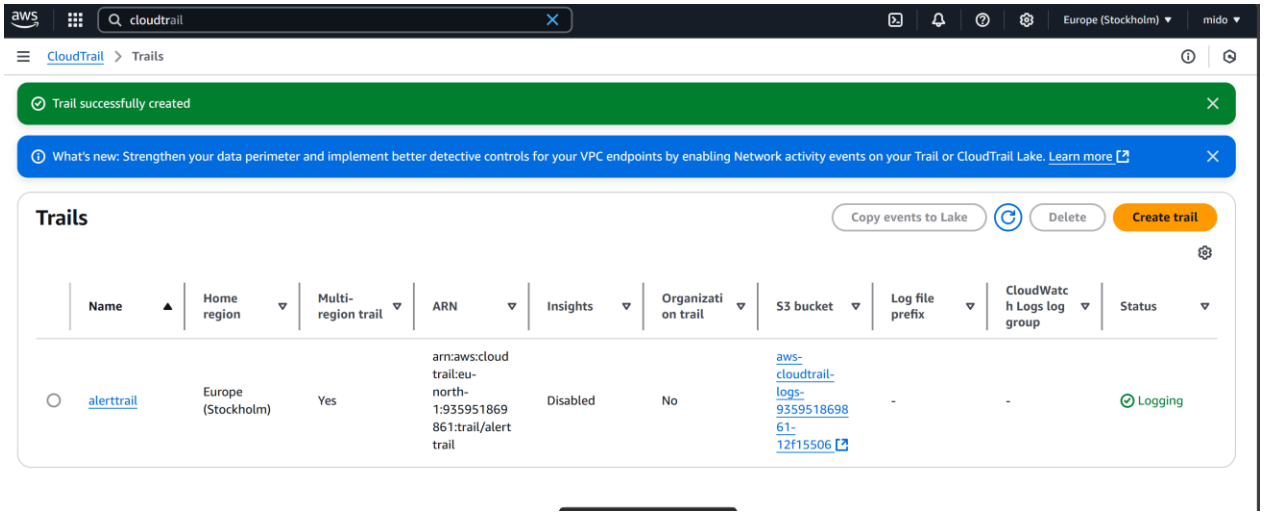
Project Steps

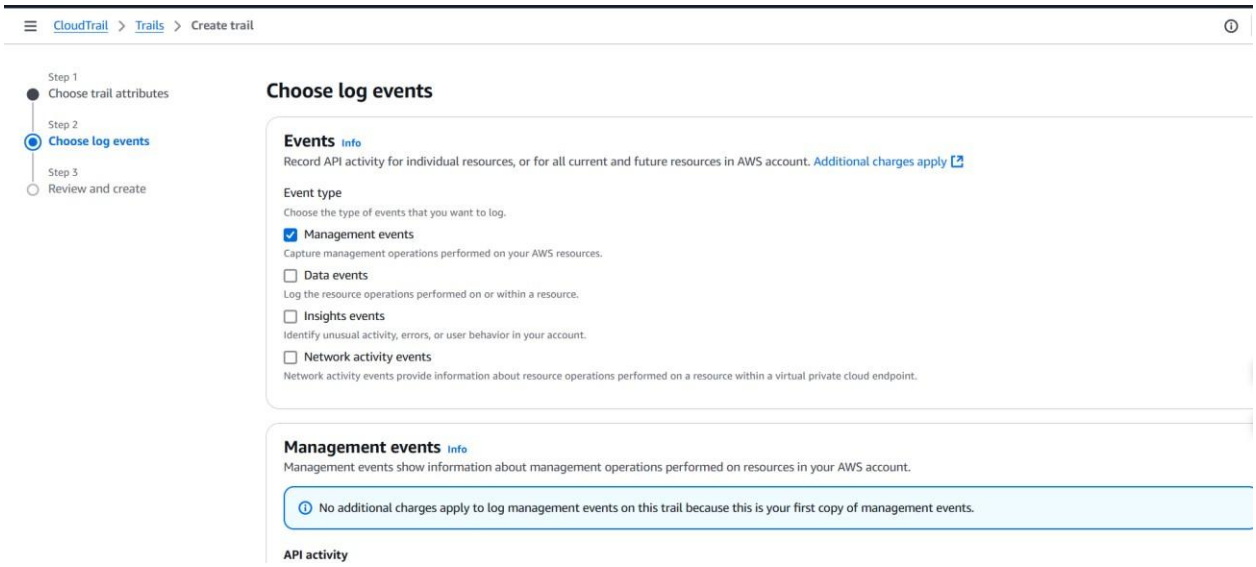
1. AWS Free Tier Account Setup

- Created a Free Tier AWS account on aws.amazon.com.
- Configured billing information and identity verification.
- Selected "Basic Support – Free" for cost efficiency.
- 

2. Configured AWS CloudTrail

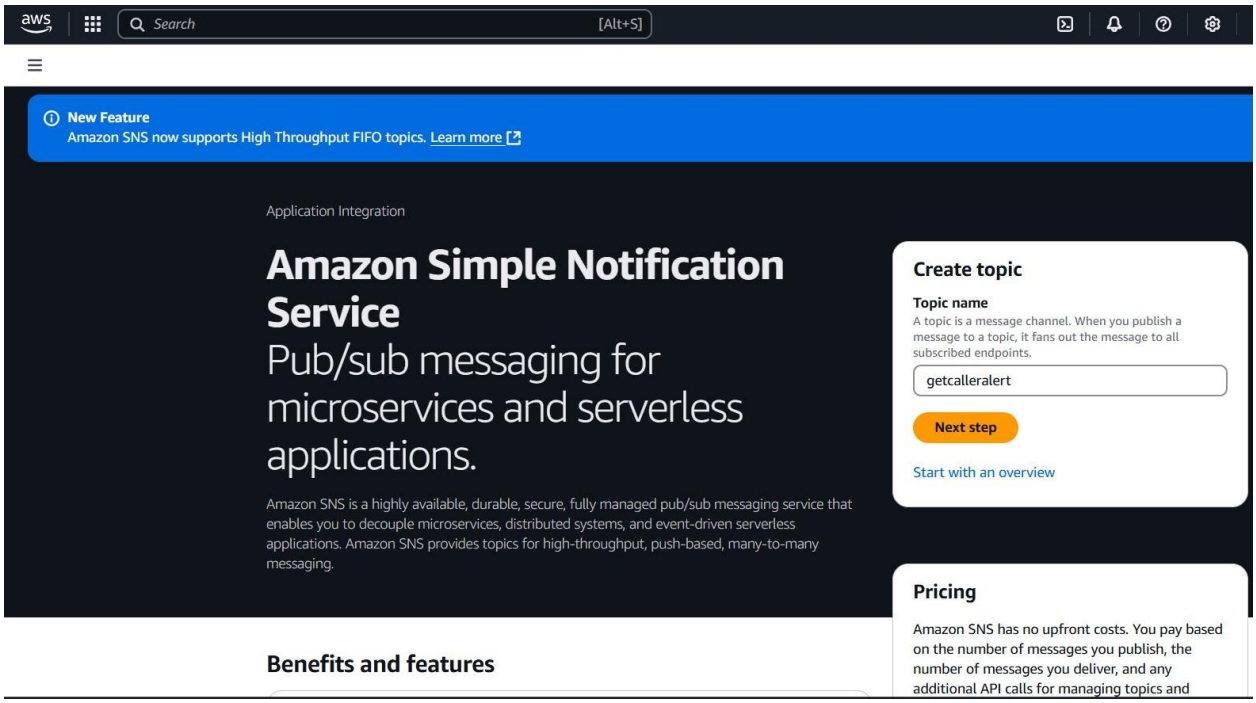
- Accessed **CloudTrail** in the AWS Management Console.
- Created a new trail named MyCloudTrail.
- Enabled **Management Event Logging** for all read/write events.
- Configured logs to be stored in an **S3 bucket** (my-cloudtrail-logs-uniqueid).
- Enabled **multi-region logging** for full coverage.




CloudTrail > Trails > Create trail
Step 1
Choose trail attributes
Step 2
Choose log events
Step 3
Review and create
Choose log events
Events Info
Record API activity for individual resources, or for all current and future resources in AWS account. Additional charges apply
Event type
Choose the type of events that you want to log.
Management events
Capture management operations performed on your AWS resources.
Data events
Log the resource operations performed on or within a resource.
Insights events
Identify unusual activity, errors, or user behavior in your account.
Network activity events
Network activity events provide information about resource operations performed on a resource within a virtual private cloud endpoint.
Management events Info
Management events show information about management operations performed on resources in your AWS account.
No additional charges apply to log management events on this trail because this is your first copy of management events.
API activity

3. Set Up SNS for Email Alerts

- Created an **SNS topic** named APICallAlert.
- Subscribed an email address to receive notifications.
- Confirmed the email subscription via the AWS verification link.
- Linked **CloudTrail** to SNS to trigger notifications when specific events occur.

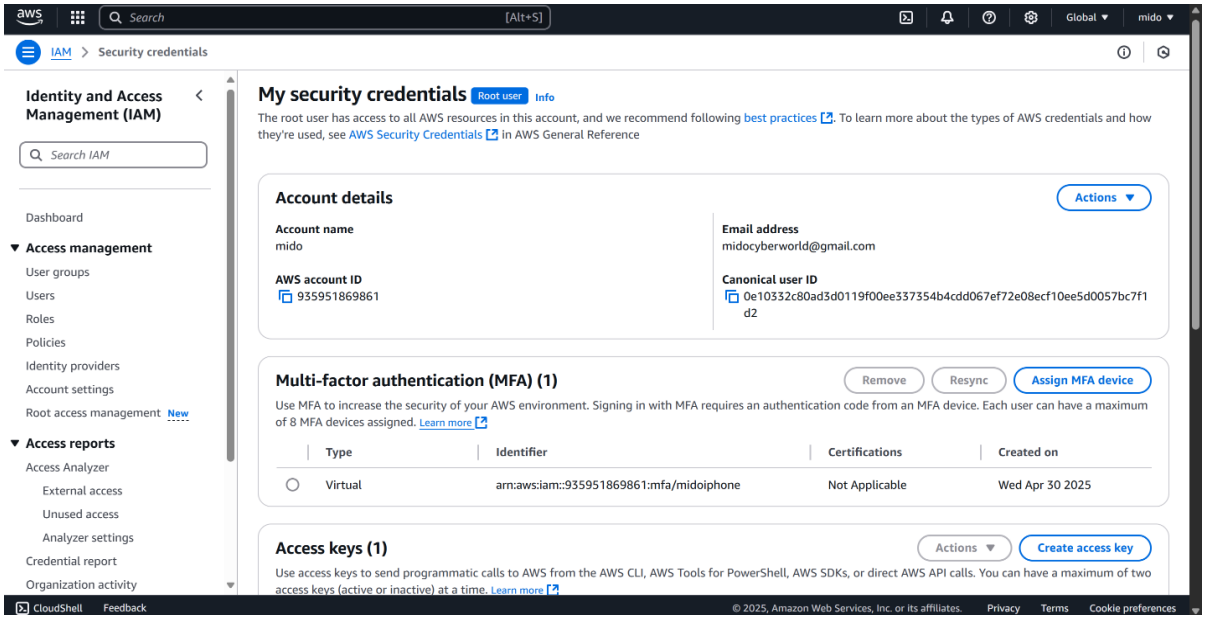

aws
Search
[Alt+S]
New Feature
Amazon SNS now supports High Throughput FIFO topics. Learn more
Application Integration
Amazon Simple Notification Service
Pub/sub messaging for microservices and serverless applications.
Amazon SNS is a highly available, durable, secure, fully managed pub/sub messaging service that enables you to decouple microservices, distributed systems, and event-driven serverless applications. Amazon SNS provides topics for high-throughput, push-based, many-to-many messaging.
Create topic
Topic name
A topic is a message channel. When you publish a message to a topic, it fans out the message to all subscribed endpoints.
getcalleralert
Next step
Start with an overview
Pricing
Amazon SNS has no upfront costs. You pay based on the number of messages you publish, the number of messages you deliver, and any additional API calls for managing topics and
Benefits and features

4. Configuring EventBridge for Root API Call Alerts

- Created an **EventBridge Rule** named GetCallerIdentityAlert.
- Defined a **custom event pattern** to detect GetCallerIdentity API calls from the root user:

```
{
  "source": ["aws.sts"],
  "detail-type": ["AWS API Call via CloudTrail"],
  "detail": {
    "eventSource": ["sts.amazonaws.com"],
    "eventName": ["GetCallerIdentity"],
    "userIdentity": {
      "type": ["Root"]
    }
  }
}
```

- Configured **SNS** as the target for this rule to send notifications.
- Allowed EventBridge to create an IAM role for permissions.


Search
[Alt+S]
Global
mido
IAM
Security credentials
Identity and Access Management (IAM)
Search IAM
Dashboard
Access management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management New
Access reports
Access Analyzer
External access
Unused access
Analyzer settings
Credential report
Organization activity
My security credentials
Root user
Info
The root user has access to all AWS resources in this account, and we recommend following best practices. To learn more about the types of AWS credentials and how they're used, see AWS Security Credentials in AWS General Reference
Account details
Actions
Account name
mido
Email address
midocyberworld@gmail.com
AWS account ID
935951869861
Canonical user ID
0e10332c80ad3d0119f00ee337354b4cdd067ef72e08ecf10ee5d0057bc7f1d2
Multi-factor authentication (MFA) (1)
Remove
Resync
Assign MFA device
Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. Learn more
Type
Identifier
Certifications
Created on
Virtual
arn:aws:iam::935951869861:mfa/midoiphone
Not Applicable
Wed Apr 30 2025
Access keys (1)
Actions
Create access key
Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. Learn more
CloudShell
Feedback
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Privacy
Terms
Cookie preferences

5. Testing the Setup via kali linux CLI

- Installed and configured **kali linux** with root credentials.
- Ran the following command to simulate an API call:

  aws sts get-caller-identity

- Verified that the API call was logged in **CloudTrail**.
- Checked the SNS email inbox for the alert notification.
- 

Results & Key Takeaways

- The system successfully detected and alerted on high-privilege root account API usage in near real-time.
- Verified the integration of **CloudTrail**, **SNS**, and **EventBridge** for a responsive and automated security monitoring setup.
- Demonstrated how AWS native tools can be combined to build a lightweight but effective **intrusion detection mechanism**.
- Gained practical experience in event-driven AWS architecture and real-world security best practices.
- Ensured alerts were functional, timely, and clearly communicated via email, providing a foundation for more advanced security workflows in future deployments.

Conclusion

This project illustrates a practical approach to implementing proactive security monitoring within AWS. By leveraging core AWS services in an integrated fashion, it's possible to create a cost-effective and efficient alerting system that identifies potentially malicious activity, such as unauthorized access via the root account. The skills demonstrated here are directly applicable to securing real-world cloud environments, especially for small to mid-size deployments prioritizing visibility and control.